

Objectives

This week, the main goal was to determine the most appropriate machine learning approach for predicting flood risk in Albay based on SYNOP station rainfall data. Specifically, we aimed to:

- Select a suitable algorithm from three candidate approaches: Prediction (regression), Clustering, and Classification.
- Implement the chosen algorithm in Python using real meteorological data from Station 98444 (Legazpi, Albay) and six surrounding stations.
- Fine-tune the model to improve its accuracy and reliability for practical flood risk assessment.

Tasks Completed

1. Choosing an Algorithm

After evaluating the three approaches, the group decided to go with Classification as the algorithm for this project. Three classifiers were then selected for comparison and implementation.

The three classifiers implemented were:

- Random Forest Classifier — selected as the primary model for its robustness with tabular data, ability to handle missing values through imputation, and built-in feature importance ranking.
- Gradient Boosting Classifier — trained as a secondary model for comparison.
- Logistic Regression — included as a linear baseline.

2. Implementation of Algorithm in Code

The implementation was organized into a structured Python pipeline (`flood_risk_predictor.py`) with the following stages:

- Data Loading — All CSV files from the `clean_dataset/` folder are loaded and parsed. Datetime columns are unified, numeric fields are coerced, and each file is keyed by its station ID.
- Feature Engineering — For Station 98444 (Albay/Legazpi), the script generates rolling rainfall windows at 12h, 24h, and 48h intervals. Lagged rainfall features (3h, 6h, 12h lookback) and pressure tendency columns were also derived. Each of the six supporting stations contributes its own individual rainfall and pressure features so the model can learn station-specific upstream signals.
- Label Generation — Flood risk is assigned using a 4-class scheme based on 24-hour accumulated rainfall thresholds aligned to Philippine DOST-PAGASA standards: LOW (< 15 mm), MODERATE (15-50 mm), HIGH (50-100 mm), and EXTREME (> 100 mm).

- **Model Training** — The dataset was split 80/20 for training and testing with stratification across risk classes. Pipelines were built using scikit-learn, with median imputation and standard scaling where needed. Models were evaluated on Accuracy, F1 Score (weighted), Precision, Recall, and AUC-OvR. A 5-fold cross-validation was also performed on the Random Forest.
- **Outlier / Extreme Event Detection** — A separate detection module flags extreme rainfall events using both IQR fencing ($Q3 + 1.5 \times IQR$) and Z-score thresholding ($|z| > 3.0$). Events are further classified into 'extreme' and 'catastrophic' tiers.

3. Fine-Tuning the Algorithm

Several adjustments were made to improve model performance and address issues that came up during initial runs:

- **Class Imbalance** — Because LOW-risk conditions dominate the dataset by a wide margin, the Random Forest and Logistic Regression were configured with `class_weight='balanced'` to prevent the model from simply predicting LOW for every observation.
- **Hyperparameter Adjustments** — The Random Forest was set to 200 trees with a max depth of 12 and `min_samples_leaf` of 5 to reduce overfitting. Gradient Boosting used a learning rate of 0.1 with 100 estimators and max depth of 5.
- **Feature Expansion** — Rather than using only Albay station data, per-station rolling windows were generated for all six supporting stations. This gave the model directional signals (e.g., Catanduanes rainfall as a typhoon approach indicator) that a purely local model would miss.
- **Validation** — 5-fold cross-validation on the Random Forest was added to assess generalization beyond the single train-test split.

Tasks to be Completed

We are planning to do the tasks that will be assigned the following week.

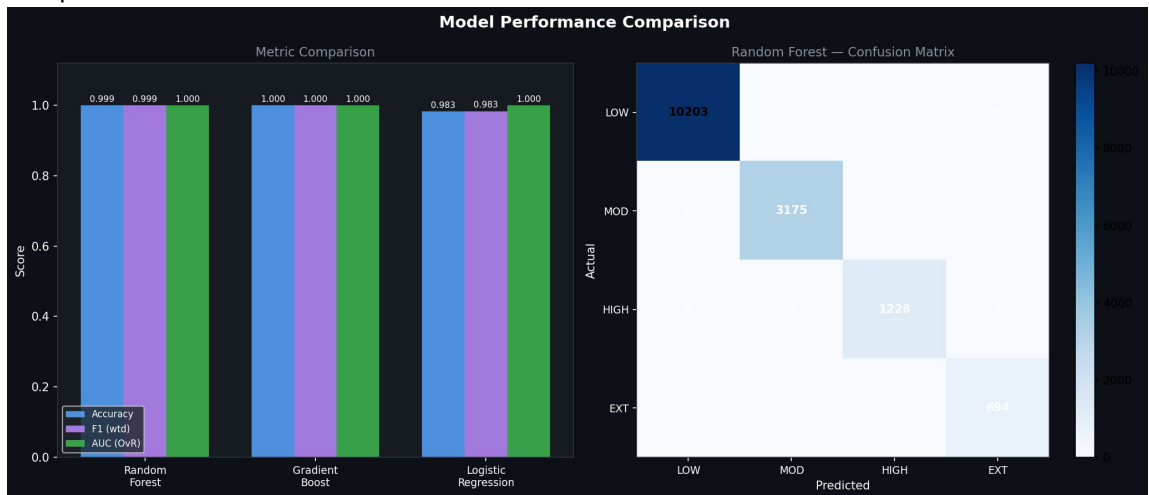
Results and Analysis

The pipeline was run on the full clean dataset covering all seven SYNOP stations from 2000 to 2026. The primary station, 98444 (Albay/Legazpi), contributed 76,544 records. The six supporting stations added further context: Camarines Norte (98440) with 76,899 records, Catanduanes (98446) with 76,283, Infanta (98434) with 75,647, Masbate (98543) with 75,956, Romblon (98536) with 73,888, and Tayabas (98427) with 75,034. Each supporting station's rainfall, pressure, and humidity readings were incorporated as individual features in the model alongside the Albay data. For the primary station, the risk label breakdown was 51,057 LOW, 15,877 MODERATE, 6,140 HIGH, and 3,470 EXTREME observations — confirming the class imbalance that motivated the balanced weighting during fine-tuning.

- All three models performed remarkably well. Gradient Boosting achieved the highest scores with 99.97% accuracy and a perfect 1.000 AUC, followed closely by Random Forest at 99.94% accuracy and 0.9999 AUC. Logistic Regression performed well too at 98.32% accuracy, though noticeably behind the tree-based models. The confusion matrix for Random Forest showed

near-perfect classification across all four risk classes — 10,203 LOW, 3,175 MODERATE, 1,228 HIGH, and 694 EXTREME correctly predicted with virtually no misclassifications.

- Feature importance analysis from the Random Forest showed that rain_24h was by far the dominant predictor at 0.4530, followed by rain_48h (0.1785) and rain_12h (0.1043). Beyond the Albay-local features, the supporting stations each contributed their own signals. Catanduanes (98446) ranked highest among them with rain24h_98446 at 0.0311 and rain12h_98446 at 0.0118, reflecting its role as an upstream typhoon indicator. Camarines Norte (98440) followed with rain24h_98440 at 0.0199, and Masbate (98543) at 0.0183. Romblon (98536) and Tayabas (98427) had smaller but still present contributions at 0.0078 and 0.0073 respectively. This confirms that the multi-station design was worthwhile — the model drew on regional rainfall patterns across all six surrounding stations, not just local Albay readings.
- The outlier detection module identified 17,026 extreme rainfall events across the dataset. The timeline shows persistent extreme events throughout the 2000-2026 period, with the most intense spikes reaching nearly 1,000 mm per 3-hour window around 2019-2020. The Z-score plot confirms several catastrophic outliers well above the 3-sigma threshold, with the highest recorded Z-score exceeding 70.
- The risk distribution shows that LOW conditions account for 66.7% of all observations, with MODERATE at 20.7%, HIGH at 8.0%, and EXTREME at 4.5%. The year-by-month heatmap reveals that flood risk is highest during the typhoon season months (August through November), with some years like 2005 and 2008 showing more intense and prolonged elevated risk periods compared to others.



Albay Flood Risk Prediction — Summary Dashboard

99.9%

Accuracy

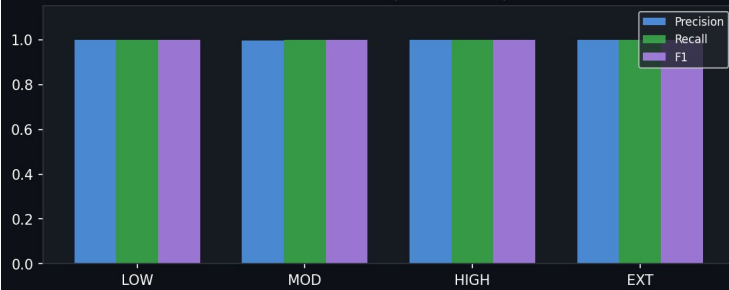
99.9%

F1 Score

0.9999

AUC (OvR)

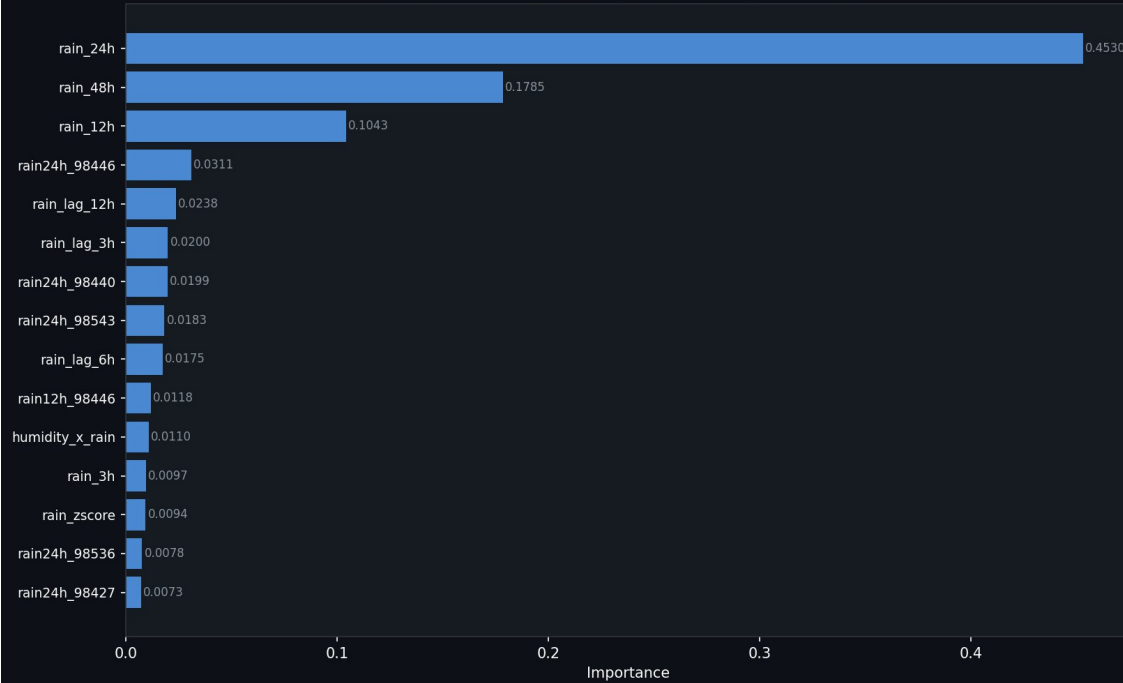
Per-Class Metrics (Random Forest)



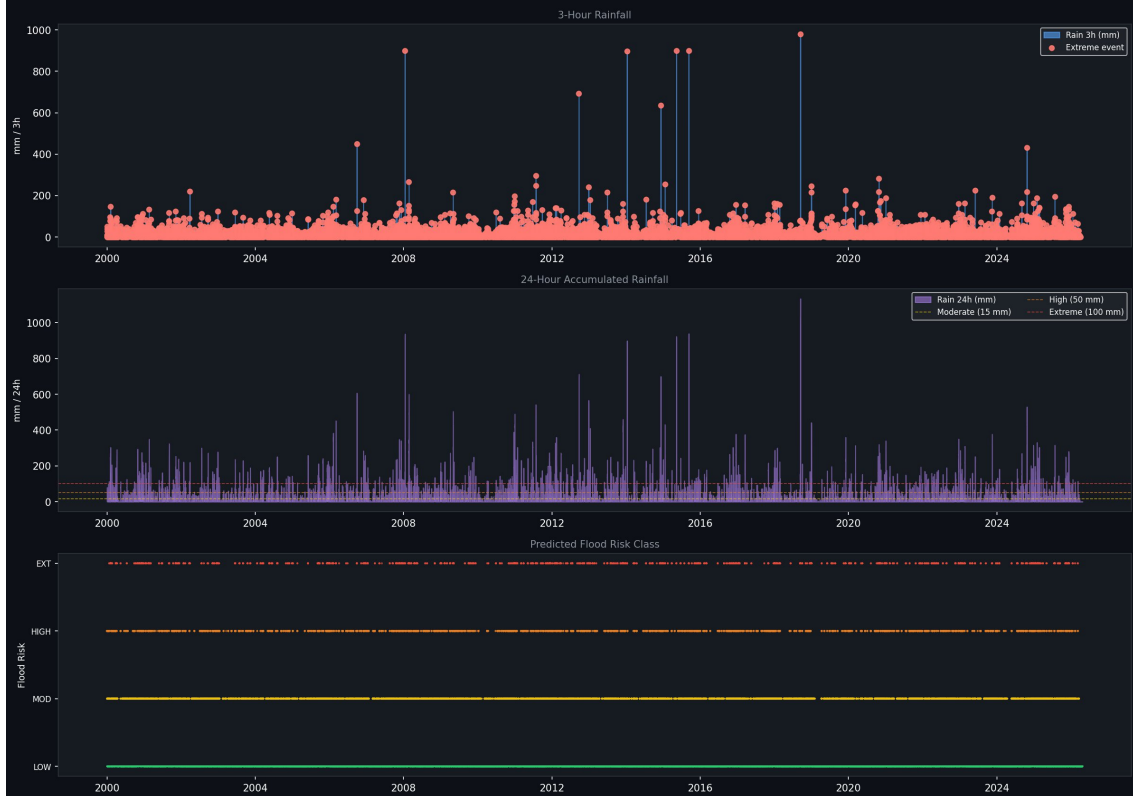
Records per Risk Class



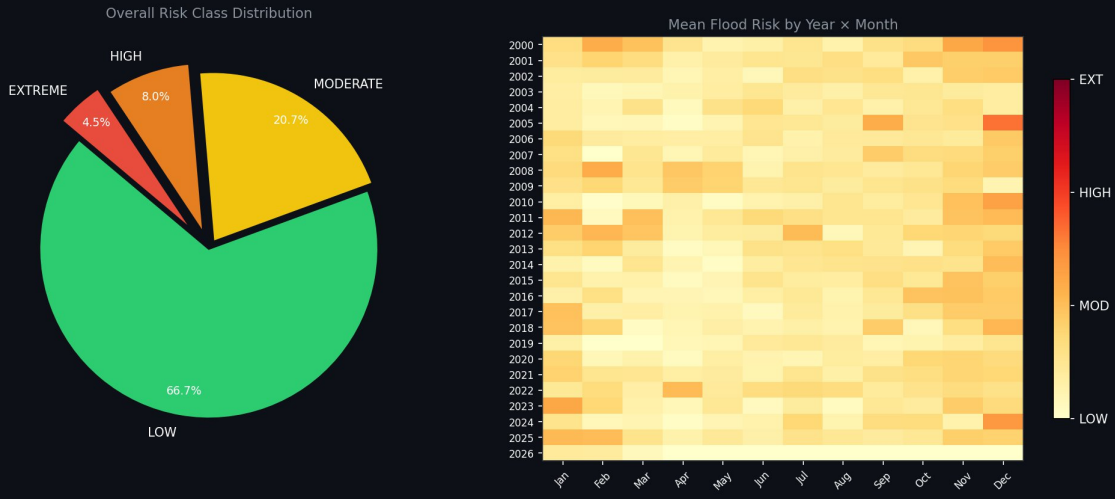
Random Forest — Top Feature Importances

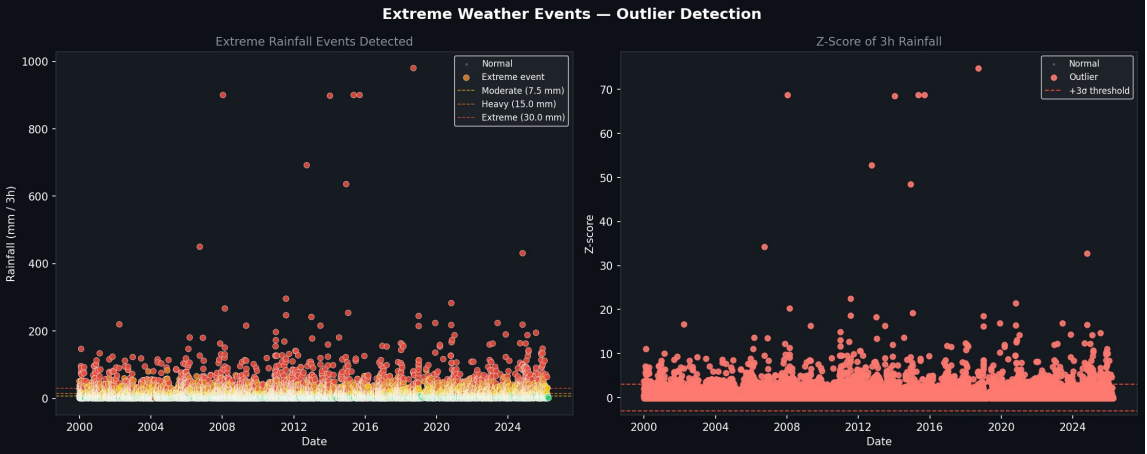


Albay (Station 98444) — Rainfall Timeline & Flood Risk



Flood Risk Distribution — Albay





datetime	station_id	rain_3h	rain_12h	rain_24h	rain_48h	temp	pressure	humidity	flood_risk	flood_risk_extreme_e	extreme_tirain_zscore
1/1/2000	98444	51.0	51.0	51.0	51.0	51.0	24.8	1011.7	98.8	2 HIGH True	catastrophic 3.7018000000000000
1/1/2000 3:00	98444	3.0	54.0	54.0	54.0	26.2	1011.1	92.6	2 HIGH True	extreme	#####
1/1/2000 6:00	98444	9.0	63.0	63.0	63.0	28.0	1008.5	86.8	2 HIGH True	extreme	#####
1/1/2000 9:00	98444	0.0	63.0	63.0	63.0	28.2	1008.7	84.3	2 HIGH False	normal	#####
1/1/2000 12:00	98444	0.0	12.0	63.0	63.0	27.1	1011.2	86.2	2 HIGH False	normal	#####
1/1/2000 15:00	98444	0.0	9.0	63.0	63.0	25.4	1011.5	93.6	2 HIGH False	normal	#####
1/1/2000 18:00	98444	0.0	0.0	63.0	63.0	26.2	1009.4	82.1	2 HIGH False	normal	#####
1/2/2000	98444	10.0	10.0	73.0	73.0	25.7	1011.4		2 HIGH True	extreme	0.5615530000000000
1/2/2000 3:00	98444	0.1	10.1	22.1	73.1	29.2	1011.7	75.4	1 MODERATE False	normal	#####
1/2/2000 6:00	98444	28.0	38.1	47.1	101.1	25.5	1009.6	95.3	1 MODERATE True	extreme	1.9401980000000000
1/2/2000 9:00	98444	0.1	38.2	38.2	101.2	26.8	1009.0	88.3	1 MODERATE False	normal	#####
1/2/2000 12:00	98444	0.0	#####	38.2	101.2	25.6	1011.0	90.9	1 MODERATE False	normal	#####
1/2/2000 15:00	98444	0.0	28.1	38.2	101.2	25.0	1011.7	90.8	1 MODERATE False	normal	#####
1/2/2000 18:00	98444	0.0	0.1	38.2	101.2	25.2	1010.0	90.9	1 MODERATE False	normal	#####
1/2/2000 21:00	98444	10.0	10.0	48.2	111.2	24.0	1009.7	97.0	1 MODERATE True	extreme	0.5615530000000000
1/3/2000	98444	38.0	48.0	76.2	149.2	24.8	1011.8	93.1	2 HIGH True	catastrophic	2.7061120000000000
1/3/2000 3:00	98444	4.0	52.0	80.10000000000000	102.2	25.8	1012.0	91.4	2 HIGH True	extreme	0.1020050000000000
1/3/2000 6:00	98444	4.0	56.0	56.1	103.2	27.0	1009.7	85.2	2 HIGH True	extreme	0.1020050000000000
1/3/2000 9:00	98444	0.4	46.4	56.4	#####	26.4	1009.6	89.3	2 HIGH False	normal	#####
1/3/2000 12:00	98444	0.3	8.7	56.7	94.9	25.0	1011.4	89.2	2 HIGH False	normal	#####
1/4/2000	98444	4.0	8.7	60.7	98.9	26.0	1012.4	83.6	2 HIGH True	extreme	0.1020050000000000
1/4/2000 3:00	98444	0.0	4.7	60.7	98.9	28.9	1012.7	74.0	2 HIGH False	normal	#####
1/4/2000 6:00	98444	0.1	#####	50.8	99.0	29.3	1010.1	73.6	2 HIGH False	normal	#####
1/4/2000 9:00	98444	1.0	5.1	13.8	90.0	26.4	1009.7	91.5	0 LOW True	extreme	#####

model	accuracy	f1_weighted	precision_w	recall_weight	auc_ovr
Random Fo	0.9994	0.9994	0.9994	0.9994	0.9999
Gradient Bo	0.9997	0.9997	0.9997	0.9997	1.0
Logistic Reg	0.9832	0.9834	0.9838	0.9832	0.9997

datetime	station_id	rain_3h	rain_zscore	flood_risk_l	extreme_tier	temp	pressure	humidity
1/1/2000	98444	51.0	3.7018000000000000	HIGH	catastrophic	24.8	1011.7	98.8
1/1/2000 3:00	98444	3.0	0.025413000000000000	HIGH	extreme	26.2	1011.1	92.6
1/1/2000 6:00	98444	9.0	0.484962000000000000	HIGH	extreme	28.0	1008.5	86.8
1/2/2000	98444	10.0	0.561553000000000000	HIGH	extreme	25.7	1011.4	
1/2/2000 6:00	98444	28.0	1.940198000000000000	MODERATE	extreme	25.5	1009.6	95.3
1/2/2000 21:00	98444	10.0	0.561553000000000000	MODERATE	extreme	24.0	1009.7	97.0
1/3/2000	98444	38.0	2.706112000000000000	HIGH	catastrophic	24.8	1011.8	93.1
1/3/2000 3:00	98444	4.0	0.102005000000000000	HIGH	extreme	25.8	1012.0	91.4
1/3/2000 6:00	98444	4.0	0.102005000000000000	HIGH	extreme	27.0	1009.7	85.2
1/4/2000	98444	4.0	0.102005000000000000	HIGH	extreme	26.0	1012.4	83.6
1/4/2000 9:00	98444	1.0	-00.127769000000000000	LOW	extreme	26.4	1009.7	91.5
1/4/2000 12:00	98444	1.0	-00.127769000000000000	LOW	extreme	25.1	1012.6	89.8
1/6/2000	98444	2.0	-00.051178000000000000	LOW	extreme	25.2	1011.6	95.3
1/6/2000 6:00	98444	1.0	-00.127769000000000000	LOW	extreme	28.6	1009.0	80.0
1/7/2000	98444	1.0	-00.127769000000000000	LOW	extreme	23.6	1012.3	97.0
1/10/2000	98444	20.0	1.327467000000000000	MODERATE	extreme	24.2	1013.6	94.2
1/10/2000 18:00	98444	1.0	-00.127769000000000000	MODERATE	extreme	26.4	1012.9	83.6
1/11/2000	98444	1.0	-00.127769000000000000	MODERATE	extreme	26.4	1012.9	83.6
1/12/2000 15:00	98444	2.0	-00.051178000000000000	LOW	extreme	25.4	1009.7	93.6
1/12/2000 18:00	98444	4.0	0.102005000000000000	LOW	extreme	25.2	1008.3	95.3
1/12/2000 21:00	98444	4.0	0.102005000000000000	LOW	extreme	24.5	1008.0	98.8
1/13/2000	98444	8.0	0.408370000000000000	MODERATE	extreme	24.8	1009.6	98.8
1/13/2000 3:00	98444	15.0	0.944510000000000000	MODERATE	extreme	25.7	1010.3	93.1

Challenges and Solutions

Challenge 1: Class Imbalance in the Risk Labels

When we first ran the model, it was predicting LOW risk for almost everything and still getting a deceptively high accuracy. We realized that most of the dataset naturally falls under LOW risk conditions, which meant the model was just taking the easy route.

Solution: We added `class_weight='balanced'` to both the Random Forest and Logistic Regression classifiers, which tells scikit-learn to weight minority classes (HIGH and EXTREME) more heavily during training. We also added `stratify=y` to the `train_test_split` call so that each risk class was proportionally represented in both the training and test sets.

Challenge 2: Aligning Multiple Station Timestamps

The six supporting stations do not always share the same timestamps as the main Albay station. When we tried to merge them directly, the result was a lot of NaN values or misaligned rows that broke the rolling window calculations.

Solution: In the `engineer_features` function, each supporting station's data is resampled to a uniform 3-hour grid using `.resample('3h').mean()` and then re-indexed to match the Albay station's `datetime` index. This ensures all station features are properly aligned before any rolling or lag operations are applied.

Challenge 3: Feature Matrix Growing Too Wide

Once we added per-station features for all six supporting stations (each with their own rain rolling windows, pressure tendency, and humidity), the feature matrix expanded significantly. We were worried this would cause overfitting or slow down training.

Solution: The Random Forest's `max_depth=12` and `min_samples_leaf=5` parameters naturally constrain model complexity even with a wide feature set. We also used the feature importance plot to verify that the model was not relying on noise columns. The 5-fold cross-validation further confirmed that the model was generalizing and not just memorizing the training data.

Conclusion

All three tasks for this week were completed. The group selected Classification as the algorithm, built out a full multi-station prediction pipeline, and fine-tuned the models. Running the pipeline on the complete dataset confirmed that the approach works well — all three classifiers achieved accuracy above 98%, with Gradient Boosting and Random Forest both clearing 99.9%.

One clear takeaway from the results is that accumulated rainfall over longer windows (24h and 48h) is far more predictive than instantaneous readings, accounting for over 63% of the model's decision-making by feature importance. The inclusion of neighboring station data — particularly Catanduanes (98446) — also contributed meaningfully, validating the multi-station design of the script. The outlier detection further added practical value by surfacing historically extreme events that a plain classifier would not have flagged on its own.